

CHE 201 Fall 2019 HW 3

Mark Hurley

TOTAL POINTS

100 / 100

QUESTION 1

11 15 / 15

✓ - **0 pts** Correct

- **7 pts** b. was missing
- **2 pts** $W = -2\text{KJ}$; $\Delta PE = 2\text{KJ}$
- **7 pts** b. incorrect

QUESTION 2

22 15 / 15

✓ - **0 pts** Correct

- **3 pts** unit conv for Rotation
- **5 pts** c. was incorrect.

QUESTION 3

33 20 / 20

✓ - **0 pts** Correct

- **5 pts** Wrong solution for work (calculation and/or units conversion error). Correct answer is $Work = 30.794 \text{ Btu/lb}$
- **5 pts** Correct equation, but wrong solution for T_2 . Correct answer is 819.9 oR
- **2.5 pts** $\Delta u = -W = -30.79 \text{ Btu/lb}$; Correct answer for T_2 is 819.9 oR
- **2.5 pts** Missing energy balance
- **20 pts** Wrong solution
- **5 pts** Did not state the postulates applied to solve the problem

QUESTION 4

44 15 / 15

✓ - **0 pts** Correct

- **5 pts** Wrong equation for coefficient of performance. The correct answer is 3
- **15 pts** Wrong answer. $Q_{in} = 0.45 \text{ kw}$ and $\beta = 3$

QUESTION 5

55 15 / 15

✓ - **0 pts** Correct

- **15 pts** Wrong or missing answer
 - **4 pts** $V_a = 9.94 \times 10^{-3} \text{ m}^3/\text{kg}$
 - **4 pts** $V_b = 0.9992 \times 10^{-3} \text{ m}^3/\text{kg}$
 - **4 pts** $V_c = \text{aprox } 1.0078 \times 10^{-3} \text{ m}^3/\text{kg}$ ($= V_f \text{ sat at } 40 \text{ oC}$)
 - **3 pts** Wrong or missing sketch of T-v diagram
 - **1 pts** b and c should be at the same temperature (sketch)
 - **2 pts** Should not extrapolate the table limits. $V_c = \text{aprox } 1.0078 \times 10^{-3} \text{ m}^3/\text{kg}$ ($= v_{fsat} \text{ at } 40 \text{ oC}$)
 - **1 pts** Point c should be outside the L-V region (sketch)
 - **1 pts** point b should be outside the L-V region (sketch), in the compressed liquid region
 - **2 pts** $V_a = 9.94 \times 10^{-3} \text{ m}^3/\text{kg}$ (superheated vapor - Table A-4)
 - **2 pts** V_b missing the term $\times 10^{-3}$
 - **1 pts** Point a should be above the critical point
 - **1 pts** c) P should be 2Mpa and not 20MPa in the sketch of the T-v diagram
- 🗨 Regrade based on solution sent by e-mail.

QUESTION 6

66 20 / 20

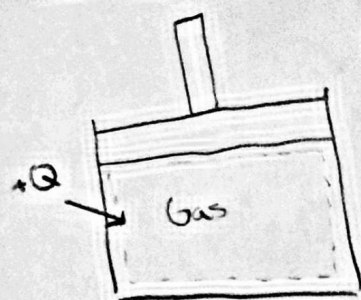
✓ - **0 pts** Correct

- **5 pts** Did not properly describe the phase-out schedule (action and deadlines) for R-22 in the US
- **5 pts** Did not list the refrigerants being used to replace R-22
- **5 pts** Did not include at least 3 references
- **3 pts** Did not properly list the references used
- **2 pts** Did not properly list the refrigerants being used to replace R-22

- **20 pts** Wrong or Missing answer
- **10 pts** Did not describe the phase-out schedule
(action and deadlines) for R-22 in the US

2.45. a.

* closed



$$V_1 = 0.1 \text{ m}^3 \quad V_2 = 0.12 \text{ m}^3$$

$$P = \text{constant.} \quad \Delta U = 0.25 \text{ kJ}$$

$$P = 2 \text{ bar} \quad P_{\text{atm}} = 1 \text{ bar}$$

$$P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$W = \int_{V_1}^{V_2} P dV = P \Delta V = 2(0.12 - 0.1) = 0.04 \left(\frac{10^5 \text{ Pa}}{1 \text{ bar}} \right) \left(\frac{1 \text{ kJ}}{10^3 \text{ N}\cdot\text{m}} \right) = 4 \text{ kJ} = W$$

$$Q - W = \Delta U + \Delta KE + \Delta PE$$

$$Q = \Delta U + W = 0.25 + 4 = 4.25 \text{ kJ} = Q$$

* P is constant

$$b. W = \int_{V_1}^{V_2} (P_{\text{atm}} - P_g) dV = 0.02 \left(\frac{10^5 \text{ Pa}}{\text{bar}} \right) \left(\frac{\text{kJ}}{10^3 \text{ J}} \right) = -2 \text{ kJ}$$

$$Q - W = \Delta U + \Delta KE + \Delta PE$$

$$\Delta PE = -W = -(-2) = 2 \text{ kJ}$$

$$2.47 \quad i = 20 \text{ amps}$$

$$V = 110$$

$$\tau = 9.07 \text{ N}\cdot\text{m}$$

$$2000 \text{ RPM} = \left(\frac{1 \text{ m}}{60 \text{ sec}} \right) \left(\frac{2\pi}{1 \text{ REV}} \right) = 66.66\pi = \omega$$

$$\text{Input power} = V \cdot i = 110 \cdot 20 = \boxed{2200 \text{ watts}} = 2.2 \text{ kW}$$

$$\text{Output power} = \tau \cdot \omega = 604.6\pi = \boxed{1900 \text{ watts}} = 1.9 \text{ kW}$$

* 0.09 per kW·hr

$$(2.2 \text{ kW}) (24) (0.09) = \boxed{\$4.752} \text{ for 24 hours run time.}$$

11 15 / 15

✓ - 0 pts Correct

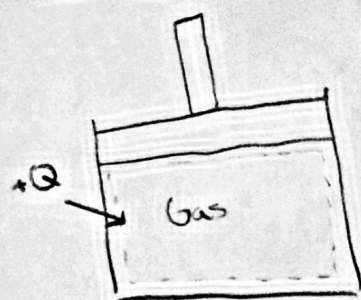
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22 15 / 15

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- 3 pts unit conv for Rotation

- 5 pts c. was incorrect.

Mark Hurley HW 3 ch 201

2.53

$$P_1 = 100 \text{ lbf/in}^2 \quad V_1 = 3.704 \text{ ft}^3/\text{lb} \quad T_1 = 1000^\circ\text{R} \quad \Delta u = 0.171$$

$$P_2 = 50 \text{ lbf/in}^2 \quad P_1 V_1^{1.4} = P_2 V_2^{1.4} \quad \text{*polytropic}$$

$$(100)(3.704)^{1.4} = (50)V_2^{1.4}$$

$$V_2^{1.4} = 12.5$$

$$V_2^{1.4 \times 0.714} = 12.5$$

$$\times 0.714$$

$$V_2 = 6.07 \text{ ft}^3/\text{lb}$$

$$W = \frac{P_2 V_2 - P_1 V_1}{1-n} = \frac{(50)(6.07) - (100)(3.704)}{1-1.4} = 167.25 \frac{\text{ft}^3 \cdot \text{lbf}}{\text{lb} \times \text{in}^2} \left(\frac{144 \text{ in}^2}{1 \text{ ft}^2} \right) \left(\frac{1 \text{ Btu}}{778 \text{ ft} \cdot \text{lbf}} \right)$$

$$30.8 \frac{\text{Btu}}{\text{lb}}$$

Adiabatic so $Q = 0 \dots$

$$\Delta Q = \Delta U + W \quad \Delta U = -W$$

$$C_v(T_2 - T_1) = -W$$

$$0.171 T_2 = 140.2$$

$$0.171(T_2 - 1000) = -30.8$$

$$T_2 = 820^\circ\text{R}$$

2.67 $P_i = 0.15 \text{ kW}$

$$Q_{\text{loss}} = 0.6 \text{ kW}$$

$$W = Q_{\text{res}} - Q_{\text{space}}$$

$$0.15 = 0.6 - Q_{\text{sp}}$$

$$Q_{\text{sp}} = 0.45 \text{ kW}$$

$$\text{Coef/per} = \frac{Q_{\text{sp}}}{W} = \frac{0.45}{0.15} = 3$$

3.7 $T_1 = 27^\circ\text{C}$

$$T_1 = 300.15^\circ\text{K}$$

$$P_{\text{gauge}} = 300 \text{ kPa}$$

$$P_{\text{atm}} = 101,000 \text{ Pa}$$

$$P_{\text{gauge}} = 367 \text{ kPa}$$

$$P_{\text{gauge}} = P_{\text{abs}} - P_{\text{atm}}$$

$$(300,000) = P_{\text{abs}_1} - (101,000) = P_{\text{abs}_1} = 401,000$$

$$(367,000) = P_{\text{abs}_2} - (101,000) = P_{\text{abs}_2} = 468,000$$

$$P_1 T_1 = P_2 T_2$$

$$(401,000)(T_2) = (468,000)(300.15) = T_2 = 350.3^\circ\text{K} - (273.15)$$

$$T_2 = 77.1^\circ\text{C}$$

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4 4 15 / 15

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- 3 pts Wrong or missing sketch of T-v diagram
- 1 pts b and c should be at the same temperature (sketch)
- 2 pts Should not extrapolate the table limits. $V_c = \text{aprox } 1.0078 \times 10^{-3} \text{ m}^3/\text{kg}$ (= v_{fsat} at 40oC)
- 1 pts Point c should be outside the L-V region (sketch)
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- 2 pts $V_a = 9.94 \times 10^{-3} \text{ m}^3/\text{kg}$ (superheated vapor - Table A-4)
- 2 pts V_b) missing the term $\times 10^{-3}$
- 1 pts Point a should be above the critical point
- 1 pts c) P should be 2Mpa and not 20MPa in the sketch of the T-v diagram

🗨 Regrade based on solution sent by e-mail.

Montreal Protocol and CFCs

Roughly 30 years ago the first reports came in that Freon, a popular refrigerant also known as refrigerant 22 was harmful to the ozone layer. This came as quite a shock and quick action was taken in order to take Freon off the market which had been a long time favorite for many applications and appliances. This leaves the question of why was freon so popular, why was it so harmful and how it was taken off the market.

Refrigeration and air conditioning are very popular products in the modern world. They rely on refrigerants who have specific properties such as a extremely low boiling point of almost 0 degrees centigrade and great structural stability to function properly(ch.ic.ac.uk). Freon is a class of chlorofluorocarbons that was the first of its kind in terms of its properties and viability for commercial use.

Chlorofluorocarbons also known as CFCs, were first synthesized in the 1890s but most were flammable and dangerous. This was until in 1930 the first safe refrigerant, Freon, was created (Coolingco). It was very quickly used in refrigeration and as a propellant in aerosol cans. In 1974 at the University of California it was first theorized that CFCs could be causing damage to the ozone layer (Facilitiesnet). Nearly 10 years later and after years of research the Montreal Protocol was passed. This was an international agreement for the phaseout of CFCs(Coolingco).

The goal projection was to be under a 10% usage level by 2015 and complete phaseout (99.5%) by Jan 1, of 2020(Bryce). This would take a large-scale operation as the refrigerant could not be vented into the atmosphere and would have to be carefully dealt with. The main driving force of this was simply halting of production of all freon using equipment. Many scrap yards will not even take items that cannot be properly proved that the refrigerant inside was dealt with correctly. Currently freon can only be procured from recycled older devices.

As of recent the types of refrigerants being used instead of CFCs they are called hydrofluorocarbons or HFCs or hydrofluoroolefins or HFOs. HFCs do not have the same ability to damage the ozone layer but can act as a potent greenhouse gas and have been used less as of 2016 and instead HFOs are beginning to be implemented even more(Bryce). Refrigerants such as R-404A, R-134a, and R-410A and R-407C that were used instead of R-22 are seem as more of an intermediate because of their potential risk as a greenhouse gas and are to be phased out as well by 2024 as ruled by the EPA(Coolingco).

Works Cited

Company, The Cooling. "What Types of Refrigerant Are Used in AC Nowadays?" Las Vegas Air Conditioning & Heating Repair, <https://www.thecoolingco.com/blog/what-types-of-refrigerant-are-used-in-ac-nowadays>.

"What You Need To Know About the HFC Refrigerant Phase-Out - Facilities Management Insights." Facilitiesnet, 10 July 2017, <https://www.facilitiesnet.com/hvac/article/What-You-Need-To-Know-About-the-HFC-Refrigerant-Phase-Out--17286>.

Bryce, et al. "What Is Freon (And Why It's In Your Air Conditioner)." Compact Appliance, 6 Jan. 2017, <https://learn.compactappliance.com/freon-air-conditioner/>.

CFCs, <http://www.ch.ic.ac.uk/rzepa/mim/environmental/html/cfc.htm>.

6 6 20 / 20

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- 5 pts Did not properly describe the phase-out schedule (action and deadlines) for R-22 in the US
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- 5 pts Did not include at least 3 references
- 3 pts Did not properly list the references used
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- 20 pts Wrong or Missing answer
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